

MOHAMMED SAMI JAITAPKAR

Graduate Mechanical Engineer | CAD • CFD • FEA • Build & Test

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PROFILE

Mechanical engineering graduate with five years of formal training: BEng (Hons) 2:1, Anglia Ruskin University (2026), built on a three-year Diploma in Mechanical Engineering (83%, Distinction). Takes designs from Inventor CAD through ANSYS computational fluid dynamics (CFD) and finite element analysis (FEA) to physical build and test. Individual dissertation used conjugate heat transfer CFD to compare Formula 1 brake cooling ducts, verified to a 0.82% grid convergence index. Coursework includes a BAE Systems live brief taken from FEA to additive manufacture and testing, and an HVAC duct redesign that cut pressure drop by 27%. Seeking graduate roles in mechanical design, simulation, building services or manufacturing.

EDUCATION

BEng (Hons) Mechanical Engineering, Upper Second Class (2:1)

2024 – 2026

Anglia Ruskin University, Chelmsford

- Dissertation: Design and Numerical Investigation of a Formula 1 Brake Cooling System (CFD and conjugate heat transfer).
- Key modules: Materials Science, Solid Mechanics & Vibration (73%); CAE & Manufacturing Project (70%); Advanced Structural Analysis & Design (67%); CFD & MATLAB/Simulink; Fluid Mechanics & Heat Transfer; Automation & Control; Advanced Engineering Mathematics; Design, Project Management & Quality Engineering.

Diploma in Mechanical Engineering, 83.13% (First Class with Distinction)

2019 – 2022

Mohammed Haji Saboo Siddik Polytechnic, Mumbai (three-year full-time technical diploma)

- Machine design, theory of machines, thermal engineering, fluid mechanics, strength of materials, engineering drawing, CAD and solid modelling, additive manufacturing. Final project: voice-based lift control system.

ENGINEERING PROJECTS

Formula 1 Brake Cooling Duct: CFD and Conjugate Heat Transfer

Individual dissertation, 2026

- Modelled three configurations in Autodesk Inventor (unducted disc, three-outlet dome duct A, single-outlet bellmouth scoop B) to find which inlet removes the most heat from a 278 mm, 24-vane carbon-carbon disc at 1,500 rpm and 20 m/s.
- Ranked designs by heat transfer coefficient using isothermal moving-wall runs in ANSYS Fluent 2025 R2 (k- ω SST, pressure-based coupled solver), then ran staged conjugate heat transfer (CHT) to resolve the temperature field inside the solid disc.
- Verified mesh independence with a three-mesh Richardson Grid Convergence Index of 0.82% on a 9.7M-cell mesh (against a published benchmark of 1.40%), supported by y+ checks and meshes built in both ANSYS Meshing and Fluent Meshing.
- Justified geometry simplifications from literature and planned parallel simulation runs across multiple lab PCs.
- Showed Duct B cooled the disc 109 K more than Duct A on 65% less mass flow, traced to an 88.7% higher inlet Reynolds number on a 30.2 mm hydraulic diameter; CHT resolved a 712 K peak disc temperature and exposed thermal runaway on the unducted disc.

Automotive HVAC Duct: CFD Optimisation and Cabin Thermal Model

CFD & Systems Engineering project

- Iterated the duct through six CAD-CFD loops, screening concepts in ANSYS Discovery before detailed ANSYS Fluent analysis on a poly-hexcore mesh, and reworked the duct routing on each loop.
- Reduced system pressure drop by 27% (~199 Pa to ~145 Pa, final Design 4.2) and balanced the outlet mass-flow split to near-equal (~50/50) between the two vents.
- Identified peak velocity (~26 m/s) as the remaining limit exceedance and set it as the constraint for the next iteration.
- Coupled the CFD results to a Simulink/Simscape cabin heating model to assess system-level thermal behaviour.

SPIDR-Jet Hybrid Drone Chassis

BAE Systems live brief

- Developed a jet-powered hybrid drone chassis from CAD through four ANSYS Mechanical FEA iterations under weight and impact load cases.
- Rejected a full-lattice PLA concept that failed structural checks and selected a selective-lattice PETG chassis, cutting mass by 13.8% to 93.48 g at a peak von Mises stress of 23.4 MPa (safety factor 1.54).
- Validated manufacturability end to end with ANSYS Additive Print, Bambu Studio FDM slicing and Tinius Olsen compression testing of printed parts.
- Modelled production in Witness Horizon discrete event simulation, scaling throughput from ~42 to ~500 units a month across printer-count and shift scenarios.
- Presented the final design in a BAE Systems-facing pitch and a ~4,000-word technical report.

Leaf Spring: FEA, Materials Selection and Lab Validation

Module mark 73%

- Modelled steel and CFRP leaf springs in Autodesk Inventor and analysed stiffness in ANSYS against the design target.
- Showed CFRP met the target stiffness at ~80% lower density; checked FEA against laboratory tensile and vibration tests.

NREL 5MW Wind Turbine Blade Fatigue Life

Individual report

- Analysed blade loading in QBlade (lifting line theory with the QFEM structural model) across six operating points.
- Applied Weibull wind loading, Goodman-corrected S-N curves and Miner's rule in Excel to identify the fatigue-governing case and estimate blade life.

Steam Pipe: Fracture Mechanics and Material Selection

Group project

- Compared EN8 and P91 steel for a steam pipe using Harrison's criterion, delivered as a group presentation.

Mechatronic Strawberry Cutter: Working Prototype

Design, build and test

- Designed the mechanism in Autodesk Inventor, the circuit in Multisim and the control logic on Arduino.
- Manufactured, assembled and commissioned the prototype in the workshop, then tested it through full cutting cycles.

TECHNICAL SKILLS

CAD & Design: Autodesk Inventor (part and assembly modelling, 2D engineering drawings), design for manufacture and assembly (DfMA), design for additive manufacture, lattice structures, Bambu Studio (FDM print planning, PETG/PLA)

CFD & FEA: ANSYS Fluent 2025 R2, Fluent Meshing (poly-hexcore, watertight geometry), ANSYS Meshing, ANSYS Discovery, ANSYS Mechanical / Static Structural, ANSYS Additive Print, QBlade (BEM/LLT + QFEM); k- ω SST, conjugate heat transfer, MRF, mesh independence, Richardson GCI, y+ checks

Thermal & Fluids: Heat transfer coefficient analysis, internal duct flow, pressure drop and flow balancing, thermal runaway assessment, fatigue (Goodman, Miner's rule), creep (Larson-Miller), fracture mechanics, NDT selection (PAUT)

Engineering Methods: Design iteration and optimisation, verification and validation (mesh independence, FEA-to-test correlation), materials selection, design trade-off analysis, literature-based modelling assumptions

Controls & MATLAB: MATLAB, Simulink, Simscape thermal networks, transfer functions, root locus, PI/PID tuning, Bode analysis, finite difference heat conduction solver, Arduino, Multisim

Manufacturing & Testing: Workshop fabrication and assembly, prototype commissioning, FDM additive manufacturing, tensile, compression (Tinius Olsen) and vibration testing, Witness Horizon discrete event simulation

Professional: Technical report writing, industry-facing presentations, project management and quality engineering (taught modules), group project work, workshop and site health and safety, Microsoft Excel

WORK EXPERIENCE

Warehouse Operative, First Call Agency, placed at DPD UK, Chelmsford

Oct 2025 – Present

- Sort, scan, load and unload parcels on night shifts, staging next-day delivery routes to hourly throughput targets.
- Work around conveyor systems in a fast-paced depot, following site safety procedures under time pressure.

Customer Service Assistant, John Lewis Partnership, Chelmsford

Nov 2023 – Oct 2025

- Provided front-of-house customer service across multiple departments, handling customer queries face to face.
- Awarded a Gold Star by management for helpfulness.

ACHIEVEMENTS & ADDITIONAL INFORMATION

- Completed the final year of the BEng (2:1) while working night warehouse shifts.
- Portfolio: nine engineering case studies with CAD, meshes, simulation results and conclusions at mosami10.github.io.
- Builds personal software projects, including a webcam gesture-controlled app and a Next.js engineering portfolio.
- Languages: English (native or bilingual); Hindi, Marathi and Urdu (professional working); Gujarati (limited working).